PIPELINE AUDIT AND HEALTH REPORT

Services: customer support call centre

Calls pass an IVR menu, then go to the shortest of five agent queues. Callers hang up when an agent queue is full (lost calls are abandoned calls). About 22% escalate to two specialists, and 7% of those come back to an agent as callbacks. Raise the call rate to see abandonment climb.

Generated 2026-10-13 19:48 | simulated 8.0 h (warm-up 30 min) | seed 33 | 10 replications

HEALTH SCORE

81 /100

Grade B

Good: sound design with minor weaknesses worth addressing.

Critical 0 | High 2 | Medium 0 | Low 2 | Info 2

The score starts at 100 and loses points for each finding. The deductions are listed in Appendix B.

THROUGHPUT	COMPLETED	ARRIVALS LOST
49.2 units/h	382 units	6.3%

AVERAGE WIP	CYCLE TIME	BOTTLENECK
13.4 units	826 s	agent_2

Executive summary

Primary bottleneck: 'agent_2' (98% busy, 0% down, index 101). 6.3% of arrivals are lost. Run what-if to measure which lever recovers the most throughput.

What to fix first

- High (arrivals): Add capacity at the constraint or enlarge the buffer in front of it. Measured best lever: 10% faster processing at 'agent_2' (+1.1% throughput).
- High (agent_2): Add parallel capacity, speed the station up, or reduce downtime. Measured best lever: 10% faster processing at 'agent_2' (+1.1% throughput).
- Low (failure model): Add realistic MTBF and MTTR to the stations that break down in practice.
- Low (throughput): Use a longer run if you need tighter estimates.

Best measured improvement

Best lever: 10% faster processing at 'agent_2' -> +0.5 units/hour (1.1%), measured over 3 seeds with common random numbers.

Findings

Each finding names where the problem is, why it matters, and what to do. Severity reflects the impact on throughput, correctness or confidence in the results.

SEVERITY	AREA	WHERE	ISSUE AND RECOMMENDED ACTION	POINTS
High	Flow	**arrivals**	6.3% of arrivals are lost because a buffer is full. Action: Add capacity at the constraint or enlarge the buffer in front of it. Measured best lever: 10% faster processing at 'agent_2' (+1.1% throughput).	-7
High	Capacity	**agent_2**	The bottleneck is 98% utilised (busy or down). There is no slack: any disturbance directly cuts output. Action: Add parallel capacity, speed the station up, or reduce downtime. Measured best lever: 10% faster processing at 'agent_2' (+1.1% throughput).	-8
Low	Reliability	**failure model**	No machine has a failure model (MTBF/MTTR), so the results are optimistic. Action: Add realistic MTBF and MTTR to the stations that break down in practice.	-3
Low	Confidence	**throughput**	Throughput 95% CI is +-5.4% of the mean. Action: Use a longer run if you need tighter estimates.	-1
Info	Capacity	**agent_2**	Constraint of the line: 98% utilised. Action: Improvements elsewhere will not raise throughput until this station is relieved. Measured best lever: 10% faster processing at 'agent_2' (+1.1% throughput).	
Info	Flow	**agent_1**	26 units were dropped here (drop-on-full policy). Action: Intentional for walk-outs and abandoned calls; otherwise switch to blocking.	

Performance by node

Share of the measured window each node spent busy, blocked (finished part cannot leave), down (failed) and starved (waiting for work).

Legend: green = busy, amber = blocked, red = down, grey = starved or idle.

NODE	KIND	STATE MIX	BUSY %	BLOCKED %	DOWN %	AVG / MAX WIP
calls	source	-	-	-	-	-
ivr	delay		35.8	0.0	0.0	0.5 / 5
agent_1	machine		94.9	5.0	0.0	2.4 / 4
agent_2 bottleneck	machine		98.3	0.0	0.0	2.1 / 4
agent_3	machine		92.7	2.0	0.0	2.0 / 4
agent_4	machine		79.6	3.8	0.0	1.7 / 4
agent_5	machine		64.7	0.0	0.0	1.3 / 4
specialist_1	machine		73.0	0.0	0.0	1.3 / 4
specialist_2	machine		82.8	0.0	0.0	2.2 / 5
resolved	sink	-	-	-	-	-

Work in process over time

Reliability and robustness

The pipeline was simulated 10 times with different random seeds. A wide interval means the result depends strongly on luck.

METRIC	MEAN	95% CONFIDENCE	STD DEV	RELATIVE
Throughput	48.75 units/h	+- 2.63	3.68	5.4%
Average WIP	18.02 units	+- 2.49	3.49	13.8%
Cycle time	1153.84 s	+- 151.36	211.60	13.1%
Arrivals lost	8.63 %	+- 4.21	5.88	48.8%

No replication deadlocked.

Failure model

No machine has a failure model.

Single points of failure

Every station has a parallel alternative: no single station can stop the whole pipeline.

Model validity checks

CHECK	RESULT	DETAIL
Unit conservation	PASS	created 433 = completed 382 + scrapped 0 + lost 26 + in system 25
Little's Law	WARN	L = 13.38, lambda x W = 12.08 (gap 9.7%)
Run length	PASS	window = 33 x cycle time (30x or more recommended)
Warm-up	PASS	first 30 min excluded from statistics
Deadlock-free	PASS	reference run completed

What-if analysis

Each lever was applied to one of the most constrained stations and the whole pipeline re-simulated over at least 8 hours with 3 identical random seeds per scenario. Baseline throughput: 47.7 units/h.

NODE	CHANGE	UNITS/H	CHANGE	CHANGE %
agent_2	10% faster processing	48.2	**+0.5**	+1.1%
agent_1	10% faster processing	48.0	**+0.4**	+0.7%
agent_1	+2 buffer slots	47.8	**+0.1**	+0.3%
agent_3	10% faster processing	47.8	**+0.1**	+0.3%
agent_2	+2 buffer slots	47.7	**+0.0**	+0.0%
agent_3	+2 buffer slots	47.7	**+0.0**	+0.0%

Best lever: 10% faster processing at 'agent_2' -> +0.5 units/hour (1.1%), measured over 3 seeds with common random numbers.

Appendix A: node inventory

Effective parameters after node-type templates are applied. Parameters not listed use the node type's defaults.

NODE	TYPE	EFFECTIVE PARAMETERS
calls	source	interarrival=exponential(65)
ivr	delay	transit=uniform(40,20)
agent_1	machine	capacity=3, on_full=drop, process=normal(300,90)
agent_2	machine	capacity=3, on_full=drop, process=normal(300,90)
agent_3	machine	capacity=3, on_full=drop, process=normal(300,90)
agent_4	machine	capacity=3, on_full=drop, process=normal(300,90)
agent_5	machine	capacity=3, on_full=drop, process=normal(300,90)
specialist_1	machine	capacity=4, process=normal(600,180)
specialist_2	machine	capacity=4, process=normal(600,180)
resolved	sink	(defaults)

Appendix B: how the score is computed

Health score = 100 minus the sum of the penalties below, never less than 0. Grades: A 90 and above, B 80 to 89, C 70 to 79, D 60 to 69, F below 60. Penalties are capped per category so that one repeated issue cannot dominate the score. Informational findings carry no penalty.

SEVERITY	WHERE	FINDING	POINTS
High	**arrivals**	6.3% of arrivals are lost because a buffer is full.	-7
High	**agent_2**	The bottleneck is 98% utilised (busy or down). There is no slack: any disturbance directly cuts output.	-8
Low	**failure model**	No machine has a failure model (MTBF/MTTR), so the results are optimistic.	-3
Low	**throughput**	Throughput 95% CI is +-5.4% of the mean.	-1
	Total deductions		**-19**
	Health score		**81**

Method: the pipeline is simulated as a discrete-event system; the bottleneck index is utilisation (busy plus down time) with small weights for starved downstream and blocked upstream neighbours. What-if results are measured, not estimated. Confidence intervals use Student's t over independent replications.